

A new hyperbolicity wedge and a joint semicircle limit for Jensen polynomials of Riemann's ξ -function

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Abstract

Let

$$\xi\left(\frac{1}{2} + z\right) = \sum_{n \geq 0} \frac{\gamma(n)}{n!} z^{2n}, \quad J^{d,n}(X) = \sum_{j=0}^d \binom{d}{j} \gamma(n+j) X^j.$$

The Riemann hypothesis is equivalent to the hyperbolicity of $J^{d,n}$ for every $d, n \geq 0$. We prove that there is an absolute constant $K > 0$ such that

$$n^3 \log^2(n+2) \geq K d^5 \implies J^{d,n} \text{ is hyperbolic.}$$

Along every sequence with $n, d \rightarrow \infty$ in this region, the empirical measure of the naturally centered and scaled zeros also converges to Wigner's semicircle law. This gives a simultaneous degree-derivative version of the global semicircle consequence of the fixed-degree Hermite limit of Griffin, Ono, Rolin, and Zagier.

1 Introduction and statement of the result

Riemann's completed zeta-function is

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s).$$

Its functional equation implies that

$$\xi_c(z) := \xi\left(\frac{1}{2} + z\right) = \sum_{n=0}^{\infty} \frac{\gamma(n)}{n!} z^{2n}$$

is an even real entire function. The coefficients satisfy $\gamma(n) > 0$. For integers $d, n \geq 0$, define

$$J^{d,n}(X) := \sum_{j=0}^d \binom{d}{j} \gamma(n+j) X^j.$$

A real polynomial is called *hyperbolic* if all its zeros are real. Pólya's Jensen criterion [7] says that the Riemann hypothesis is equivalent to the hyperbolicity of $J^{d,n}$ for all d, n .

Our main result is the following unconditional region in the (d, n) -plane.

Theorem 1.1 (Main theorem). *There is an absolute constant $K > 0$ such that, for all integers $d \geq 1$ and $n \geq 0$ satisfying*

$$n^3 \log^2(n+2) \geq K d^5,$$

the Jensen polynomial $J^{d,n}$ has d distinct negative real zeros.

Thus hyperbolicity holds uniformly for

$$d \leq c n^{3/5} \log^{2/5}(n+2)$$

with a suitable absolute $c > 0$.

There is also a consequence for the derivative-aspect random-matrix picture. Griffin, Ono, Rolin, and Zagier proved that, for each fixed d , a centered and scaled Jensen polynomial tends to a Hermite polynomial as $n \rightarrow \infty$ [2, Theorem 3]. The semicircle law for Hermite zeros then gives a global GUE statement after taking $d \rightarrow \infty$ in a second limit. The uniform comparison used here allows the two parameters to tend to infinity together.

Theorem 1.2 (Joint semicircle limit). *Let K be as in Theorem 1.1, and let (n_ν, d_ν) be any sequence of positive integers such that*

$$n_\nu \rightarrow \infty, \quad d_\nu \rightarrow \infty, \quad n_\nu^3 \log^2(n_\nu + 2) \geq K d_\nu^5.$$

Write $\rho_{1,\nu}, \dots, \rho_{d_\nu,\nu}$ for the zeros of J^{d_ν, n_ν} and put

$$R_{1,\nu} := \frac{\gamma(n_\nu + 1)}{\gamma(n_\nu)}, \quad \lambda_{j,\nu} := \sqrt{\frac{n_\nu}{d_\nu}} (1 + R_{1,\nu} \rho_{j,\nu}).$$

Then, for every bounded continuous function $f: \mathbb{R} \rightarrow \mathbb{R}$,

$$\lim_{\nu \rightarrow \infty} \frac{1}{d_\nu} \sum_{j=1}^{d_\nu} f(\lambda_{j,\nu}) = \frac{1}{2\pi} \int_{-2}^2 f(x) \sqrt{4 - x^2} \, dx.$$

Theorem 1.2 concerns the global empirical measure. It does not assert local sine-kernel statistics, eigenvalue-spacing laws, or edge universality.

The proof separates the construction of a real-rooted comparison polynomial from the stability of its zeros under the remaining coefficient perturbation. Section 2 introduces the quotient coordinates that organize the construction, displays the three nested comparison families, and proves the abstract stability principle used at the end. The subsequent sections verify its hypotheses. The analytic input is the complex form of the saddle estimate in [3, Section 3]; the algebraic input consists of Jacobi root estimates and the real-root and logarithmic-mesh properties of finite-free multiplicative convolution [4, Propositions 2.7 and 2.17]. After the hyperbolicity proof is complete, Section 11 identifies the semicircle law for the comparison model and transfers it to the Jensen zeros.

2 Comparison coordinates and proof architecture

Fix n and normalize the coefficient sequence by

$$R_0 := 1, \quad R_j := \frac{\gamma(n+j)}{\gamma(n)}, \quad q_k := \frac{R_{k+1}^2}{R_k R_{k+2}} \quad (j \geq 1, k \geq 0). \quad (1)$$

Then

$$\frac{J^{d,n}(X)}{\gamma(n)} = \sum_{j=0}^d \binom{d}{j} R_j X^j.$$

Multiplying R_j by S^j only rescales the polynomial variable and does not change any q_k . These quotient invariants are the natural coordinates for the comparison.

We note the following, whose proof is immediate:

Lemma 2.1 (Quotient coordinates). *Let $\tilde{R}_0 = R_0 = 1$ and $\tilde{R}_1 = R_1$. If*

$$\frac{\tilde{R}_{k+1}^2}{\tilde{R}_k \tilde{R}_{k+2}} = q_k \quad (0 \leq k \leq m-2),$$

then $\tilde{R}_j = R_j$ for $0 \leq j \leq m$.

The construction passes through three nested ratio families:

$$\mathbb{L}_j(B, S) := \frac{S^j}{(B)_j}, \quad B, S > 0, \quad (2)$$

$$\mathbb{J}_j(A, B, S) := \frac{S^j (A)_j}{A^j (B)_j}, \quad A, B, S > 0, \quad (3)$$

$$\mathbb{F}_j(A, B, C, D, S) := S^j \frac{(A)_j (C)_j D^j}{A^j C^j (B)_j (D)_j}, \quad A, B, C, D, S > 0. \quad (4)$$

The Laguerre family is the limit of the Jacobi family as $A \rightarrow \infty$, while

$$\mathbb{F}_j(A, B, C, C, S) = \mathbb{J}_j(A, B, S).$$

The polynomial associated with the first family is

$$\begin{aligned} \sum_{j=0}^d \binom{d}{j} \mathbb{L}_j(B, S) X^j &= {}_1F_1(-d; B; -SX) \\ &= \frac{d!}{(B)_d} L_d^{B-1}(-SX), \end{aligned} \quad (5)$$

so it has d simple negative zeros.

The role of the three families can now be stated without their parameter calculations. The Laguerre family is scaled to match R_1 and R_2 . Its error at R_3 has a definite positive sign; that sign permits a Jacobi parameter $A > 0$ to match R_1, R_2, R_3 . The Jacobi model has a defect of the opposite sign at R_4 . A second Jacobi factor, introduced through finite-free multiplicative convolution, absorbs that defect and produces a model matching $R_0, \dots, R_4$ exactly. Lemma 2.1 reduces each matching assertion to the corresponding equations for the q_k .

It remains to explain why matching five coefficients is enough. The following proposition isolates the final argument from the special functions used to construct the model.

Proposition 2.2 (Fifth-order multiplier stability). *Let $d \geq 5$, and let*

$$p(y) = \sum_{j=0}^d p_j y^j$$

be a real polynomial with d simple positive zeros and $p(0) \neq 0$. For $r > 0$, put

$$\Omega_r := \{z \in \mathbb{C} : \text{dist}(z, [0, d]) \leq 2r\}.$$

Suppose that c is holomorphic on a neighborhood of Ω_r , that $c(0), \dots, c(d)$ are real, and that

$$c(0) = \dots = c(4) = 1, \quad c(d) > 0, \quad \sup_{\Omega_r} |c - 1| \leq \varepsilon < 16.$$

Define

$$P(y) := \sum_{j=0}^d p_j c(j) y^j.$$

If, at every critical point y of p ,

$$\left| \frac{y^k p^{(k)}(y)}{p(y)} \right| \leq r^k \quad (0 \leq k \leq d), \quad (6)$$

then P has d simple positive zeros. Moreover, if $N_Q(t)$ denotes the number of zeros of a polynomial Q in $(0, t]$, counted with multiplicity, then

$$\sup_{t>0} |N_P(t) - N_p(t)| \leq 1. \quad (7)$$

Proof. Write

$$\Delta^k c(0) := \sum_{\ell=0}^k (-1)^{k-\ell} \binom{k}{\ell} c(\ell).$$

Newton interpolation gives

$$P(y) = \sum_{k=0}^d \Delta^k c(0) \frac{y^k p^{(k)}(y)}{k!}. \quad (8)$$

Indeed, substitute $c(j) = \sum_{k=0}^j \binom{j}{k} \Delta^k c(0)$ and interchange the two finite sums.

For $k \geq 1$, repeated use of the fundamental theorem of calculus gives

$$\Delta^k c(0) = \int_{[0,1]^k} c^{(k)}(t_1 + \cdots + t_k) dt_1 \cdots dt_k.$$

Cauchy's estimate on disks of radius $\rho < 2r$ centered on $[0, d]$, followed by $\rho \uparrow 2r$, therefore yields

$$\frac{|\Delta^k c(0)|}{k!} \leq \frac{\varepsilon}{(2r)^k}. \quad (9)$$

The five matching conditions imply

$$\Delta^0 c(0) = 1, \quad \Delta^k c(0) = 0 \quad (1 \leq k \leq 4).$$

At a critical point of p , equations (8)–(9) and (6) give

$$\left| \frac{P(y)}{p(y)} - 1 \right| \leq \varepsilon \sum_{k=5}^d 2^{-k} < \frac{\varepsilon}{16} < 1.$$

Thus P and p have the same sign at every critical point. They also have the same sign at 0 and for large positive y , because $c(0) = 1$ and $c(d) > 0$. The signs of p at 0, at its $d - 1$ critical points, and at $+\infty$ alternate. Hence the intermediate value theorem gives a sign-changing zero of P in each of the d intervening intervals. These exhaust its degree, so all are simple. The zeros of p' strictly interlace those of p , so p also has one zero in each of these intervals. If t lies in one of them, p and P have the same number of zeros in all preceding intervals, and their counts inside the interval containing t differ by at most one. This proves (7). $\square$

The rest of the proof verifies the hypotheses of Proposition 2.2. The saddle estimate gives the two defect signs needed to choose the model parameters and, after exact matching through R_4 , the bound

$$\sup_{\Omega_r} |c - 1| \ll \frac{d^{5/2}}{n^{3/2} \log(n+2)}.$$

The finite-free factorization gives the required positive roots, and a perturbed Jacobi equation gives (6). The condition that the displayed error be uniformly small is exactly $d^5 \ll n^3 \log^2(n+2)$.

3 Moment representation and the Laguerre baseline

The integral representation below explains the distinguished Laguerre normalization used to identify the positive defect at R_3 .

Write

$$\Xi(t) := \xi\left(\frac{1}{2} + it\right).$$

Riemann's standard integral representation has the form

$$\Xi(t) = \int_0^\infty \Phi(u) \cos(tu) \, du, \quad \Phi(u) > 0,$$

where Φ is the kernel obtained from the Jacobi theta series; the integral and all the differentiations used below converge absolutely. Put

$$M_n := \int_0^\infty \Phi(u) u^{2n} \, du.$$

Since $\xi_c(z) = \Xi(-iz)$, comparison of the expansions of $\cosh(zu)$ and $\xi_c(z)$ gives

$$\gamma(n) = \frac{n!}{(2n)!} M_n. \tag{10}$$

Define the tilted kernel probability measure

$$d\mathbb{P}_n(u) := \frac{\Phi(u) u^{2n}}{M_n} \, du$$

and let U denote its coordinate random variable. Legendre's duplication formula

$$\Gamma(z) \Gamma\left(z + \frac{1}{2}\right) = 2^{1-2z} \sqrt{\pi} \Gamma(2z)$$

gives

$$\frac{\gamma(n+j)}{\gamma(n)} = \frac{1}{4^j (n + \frac{1}{2})_j} \mathbb{E}_n[U^{2j}]. \tag{11}$$

Consequently,

$$\begin{aligned} \frac{J^{d,n}(X)}{\gamma(n)} &= \mathbb{E}_n \left[\sum_{j=0}^d \binom{d}{j} \frac{(XU^2)^j}{4^j (n + \frac{1}{2})_j} \right] \\ &= \mathbb{E}_n \left[{}_1F_1 \left(-d; n + \frac{1}{2}; -\frac{XU^2}{4} \right) \right] \\ &= \frac{d!}{(n + \frac{1}{2})_d} \mathbb{E}_n \left[L_d^{n-\frac{1}{2}} \left(-\frac{XU^2}{4} \right) \right]. \end{aligned} \tag{12}$$

For every fixed $U > 0$, the polynomial inside the expectation in (12) has d distinct negative roots, although averaging does not in general preserve hyperbolicity.

There is, however, an exact hyperbolicity-preserving random-dilation model. Let

$$b := n + \frac{1}{2}, \quad B > b,$$

and let Z have the beta distribution $\text{Beta}(b, B - b)$. Then

$$\mathbb{E}[Z^j] = \frac{(b)_j}{(B)_j}.$$

It follows immediately that, for $a > 0$,

$$\mathbb{E}\left[{}_1F_1\left(-d; b; -\frac{aZX}{4}\right)\right] = {}_1F_1\left(-d; B; -\frac{aX}{4}\right). \quad (13)$$

This is the beta–Sonine index-raising operation for Laguerre polynomials. With $S = a/4$, it realizes the Laguerre family (2) as a scaled beta dilation of the baseline in (12). This explains why that family is the first comparison model; no factorization of the tilted kernel measure is asserted.

4 The analytic saddle and the first defect

The saddle calculation is used in three places: to determine the sign of the Laguerre error at R_3 , to determine the sign of the Jacobi error at R_4 , and to bound the fifth derivative of the final residual multiplier. We record all the required derivative information here.

Continue the moment in (10) by its absolutely convergent Mellin integral and define

$$\gamma(z) := \frac{\Gamma(z+1)}{\Gamma(2z+1)} M_z.$$

The same duplication formula gives

$$\gamma(z) = \frac{\sqrt{\pi} M_z}{4^z \Gamma(z + \frac{1}{2})}, \quad M_z := \int_0^\infty \Phi(u) u^{2z} du. \quad (14)$$

To identify this continuation with the saddle formula in [3, Section 3], set

$$F(s) := \int_1^\infty (\log t)^s t^{-3/4} \sum_{k \geq 1} e^{-\pi k^2 t} dt.$$

The continuation of the exact kernel identity in [3, (3.1)], with the normalization corrected as explained below, is

$$M_z = 2^{-2z-2} \left\{ 32 \binom{2z}{2} F(2z-2) - F(2z) \right\}. \quad (15)$$

For real $x > 3$, let L_x be the positive solution of

$$x = L_x \left(\pi e^{L_x} + \frac{3}{4} \right). \quad (16)$$

For complex x in the right half-plane, L_x denotes the branch of the holomorphic continuation used in [3, Section 3]. Put

$$Q_x := (1 + L_x)x - \frac{3}{4} L_x^2.$$

The main saddle term is

$$G_0(x) := (x+1) \log L_x + \frac{L_x}{4} - \frac{x}{L_x} - \frac{1}{2} \log Q_x. \quad (17)$$

Proposition 4.1 (Sectorial logarithmic saddle). *There are $0 < \theta < \pi/2$ and $x_0 > 0$ such that M_z is nonzero in*

$$\mathfrak{S}_\theta := \{z \in \mathbb{C} : |z| \geq x_0, |\arg z| \leq \theta\}.$$

On this sector, let $h(z) = \log M_z$ be the branch real on the positive axis. If $N = 2z - 2$, then

$$h(z) = G_0(N) + \log \left(32 \binom{2z}{2} \right) - (2z + 2) \log 2 + c_{\text{sad}} + \mathcal{R}(z), \quad (18)$$

where c_{sad} is real. For every fixed $0 < \varepsilon_0 < 1/2$ and $0 \leq r \leq 5$,

$$\mathcal{R}^{(r)}(z) = O_{\varepsilon_0, r, \theta}(|z|^{-r-1+\varepsilon_0}) \quad (19)$$

uniformly on a slightly smaller closed sector.

Here h is the exact logarithm of the continued moment, whereas G_0 is only its principal saddle contribution, evaluated at $N = 2z - 2$. The remaining explicit terms account for normalization, and $\mathcal{R}$ is the controlled analytic remainder.

Proof. Write $N = 2z - 2$ and

$$K_N := (L_N^{-1} + L_N^{-2}) N - \frac{3}{4} = \frac{Q_N}{L_N^2}.$$

The complex form of [3, (3.2)], stated in the sentence following that equation, is uniform on every fixed closed sector in $\Re z > 1$ and reads, in the normalization of this paper,

$$\begin{aligned} \gamma(z) &= \frac{e^{z-2} z^{z+1/2} L_N^N}{2^{2z-2} N^{N+1/2}} \left(\frac{2\pi}{K_N} \right)^{1/2} \exp \left(\frac{L_N}{4} - \frac{N}{L_N} + \frac{3}{4} \right) \\ &\quad \times (1 + O_{\varepsilon_0}(|z|^{-1+\varepsilon_0})). \end{aligned} \quad (20)$$

Here all powers use the branches obtained by continuation from the positive axis. The factor 2^{2z-2} incorporates a normalization that is easy to miss: with $\Lambda(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$, the coefficients in [2] are those of $(4w^2 - 1) \Lambda(\frac{1}{2} + w) = 8\xi(\frac{1}{2} + w)$. Thus the right sides of [3, (3.1)–(3.2)], when used for the coefficients defined here, are divided by 8. This affects only an additive constant after taking logarithms. It does not affect the results of [3], whose use of these coefficients in Section 3 is through ratios in which the constant cancels.

The leading factor in (20) is holomorphic and nonzero on a sufficiently small fixed sector. For large $|z|$ the relative error has modulus below $1/2$, so $\gamma(z)$ and, by (14), M_z are nonzero there. Taking the logarithm, using $K_N = Q_N/L_N^2$, and applying sectorial Stirling asymptotics to $\Gamma(z + \frac{1}{2})$ in (14) gives (18). Consolidating the remaining elementary terms into the explicit terms following $G_0(N)$ changes only $c_{\text{sad}} + O(|z|^{-1})$.

The difference between the two sides of (20), divided by its nonzero leading factor, is holomorphic. Hence so is the logarithmic remainder $\mathcal{R}$. A disk $|\zeta - z| \leq \delta|z|$ about a point in a smaller closed sector stays in the original sector, and $|\zeta| \asymp |z|$ there. Cauchy's inequalities applied to the $O(|z|^{-1+\varepsilon_0})$ remainder prove (19). $\square$

Lemma 4.2 (Signed moment-saddle derivatives). *Put*

$$\mathcal{L}_x := L_{2x-2}.$$

As $x \rightarrow +\infty$,

$$h''(x) = \frac{2}{x\mathcal{L}_x} \left(1 + O\left(\frac{1}{\mathcal{L}_x} \right) \right), \quad (21)$$

$$h'''(x) = -\frac{2}{x^2\mathcal{L}_x} \left(1 + O\left(\frac{1}{\mathcal{L}_x} \right) \right). \quad (22)$$

In addition,

$$h^{(4)}(x) = \frac{4}{x^3 \mathcal{L}_x} \left(1 + O\left(\frac{1}{\mathcal{L}_x}\right) \right). \quad (23)$$

There are absolute constants $\eta, C > 0$ such that

$$|h^{(4)}(x+w)| \leq \frac{C}{x^3 \log x}, \quad (24)$$

$$|h^{(5)}(x+w)| \leq \frac{C}{x^4 \log x} \quad (25)$$

whenever $|w| \leq \eta x$.

Proof. Differentiating the saddle equation gives

$$L'_N = \frac{L_N}{Q_N}.$$

Using this identity in G_0 gives

$$G''_0(N) = \frac{1}{Q_N} + O(N^{-2}), \quad (26)$$

$$G'''_0(N) = -\frac{1}{N^2 L_N} (1 + O(L_N^{-1})), \quad (27)$$

$$G^{(4)}_0(N) = \frac{2}{N^3 L_N} (1 + O(L_N^{-1})), \quad (28)$$

$$G^{(5)}_0(N) = O\left(\frac{1}{N^4 L_N}\right). \quad (29)$$

Indeed, $Q_N = N L_N (1 + O(L_N^{-1}))$ and $Q'_N = L_N (1 + O(L_N^{-1}))$; the last two formulas follow by further differentiation of (27).

The asserted estimates follow from Proposition 4.1 and (26)–(29); the chain rule contributes the factors 2^m . The elementary terms following $G_0(2z - 2)$ in (18), and the remainder controlled by (19), are of lower order. A sufficiently small fixed $\eta > 0$ makes every disk $|w| \leq \eta x$ lie in the sector, which proves the two uniform bounds. $\square$

Lemma 4.3 (Positive Laguerre defect at R_3). *For all sufficiently large n , $q_0 > 1$. Set*

$$B_L := (q_0 - 1)^{-1}, \quad S_L := B_L R_1, \quad R_j^{(L)} := \mathbf{L}_j(B_L, S_L).$$

Then the Laguerre model matches the first two normalized coefficients,

$$R_1^{(L)} = R_1, \quad R_2^{(L)} = R_2,$$

and predicts

$$q_1^{(L)} := \frac{(R_2^{(L)})^2}{R_1^{(L)} R_3^{(L)}} = \frac{B_L + 2}{B_L + 1} = 2 - \frac{1}{q_0}.$$

Moreover,

$$\mathcal{D}_n := \log \frac{R_3}{R_3^{(L)}} = \log \frac{q_1^{(L)}}{q_1} = \frac{2}{n^2 \mathcal{L}_n} \left(1 + O\left(\frac{1}{\mathcal{L}_n}\right) \right). \quad (30)$$

In particular, $\mathcal{D}_n > 0$ for all sufficiently large n .

Proof. Put $b = n + \frac{1}{2}$ and

$$\ell(x) := \log(1 + x^{-1}), \quad \delta_0 := h(n+2) - 2h(n+1) + h(n),$$

$$\delta_1 := h(n+3) - 2h(n+2) + h(n+1).$$

Equation (14) gives

$$\log q_0 = \ell(b) - \delta_0, \quad \log q_1 = \ell(b+1) - \delta_1. \quad (31)$$

Lemma 4.2 and the integral formulas for finite differences give

$$\delta_0 = \frac{2}{n\mathcal{L}_n} (1 + O(\mathcal{L}_n^{-1})), \quad (32)$$

$$\delta_1 - \delta_0 = -\frac{2}{n^2\mathcal{L}_n} (1 + O(\mathcal{L}_n^{-1})). \quad (33)$$

Since $\ell(b) \sim n^{-1}$, equations (31) and (32) show that $q_0 > 1$ for all sufficiently large n . Direct substitution in (2) gives $R_1^{(L)} = R_1$, $R_2^{(L)} = R_2$, and the displayed formula for $q_1^{(L)}$.

Define $\mathcal{F}(x) := \log(2 - e^{-x})$. The exact relation $q_1^{(L)} = 2 - q_0^{-1}$ and $\mathcal{F}(\ell(b)) = \ell(b+1)$ imply

$$\begin{aligned} \mathcal{D}_n &= \mathcal{F}(\ell(b) - \delta_0) - \ell(b+1) + \delta_1 \\ &= \delta_1 - \frac{b}{b+2}\delta_0 + O(\delta_0^2) \\ &= (\delta_1 - \delta_0) + \frac{2}{b+2}\delta_0 + O(\delta_0^2). \end{aligned}$$

Substitution of (32) and (33) gives (30). □

5 First correction: the three-coefficient Jacobi model

We now absorb the positive R_3 defect in Lemma 4.3. For the Jacobi family (3), the first two quotient invariants are

$$q_0^{(J)}(A, B) := \frac{A(B+1)}{B(A+1)}, \quad (34)$$

$$q_1^{(J)}(A, B) := \frac{(A+1)(B+2)}{(B+1)(A+2)}. \quad (35)$$

They satisfy

$$\frac{q_0^{(J)} - 1}{q_1^{(J)} - 1} = q_0^{(J)} \left(1 + \frac{2}{A} \right). \quad (36)$$

Consequently, solving $q_0^{(J)} = q_0$ and $q_1^{(J)} = q_1$ gives the following formulas. Put

$$T := \frac{q_0 - 1}{q_1 - 1}. \quad (37)$$

For all sufficiently large n , define

$$A_J := \frac{2q_0}{T - q_0}, \quad \frac{1}{B_J} := q_0 \left(1 + \frac{1}{A_J} \right) - 1, \quad S_J := B_J R_1. \quad (38)$$

Lemma 5.1 (Positive Jacobi parameters). *The parameters in (38) are positive and satisfy*

$$A_J \sim n\mathcal{L}_n, \quad B_J \sim n, \quad B_J - \left(n + \frac{1}{2}\right) = O\left(\frac{n}{\log n}\right). \quad (39)$$

Proof. Let $q_1^{(L)} = 2 - q_0^{-1}$. Since (31), (32), and (33) give $\log q_1 \sim n^{-1}$, we have $q_1 > 1$. On the other hand, $\mathcal{D}_n = \log(q_1^{(L)}/q_1) > 0$ gives $q_1 < q_1^{(L)}$. The identity

$$q_1^{(L)} - 1 = \frac{q_0 - 1}{q_0}$$

then gives $T > q_0$. Thus $A_J > 0$, and the remaining two parameters in (38) are positive.

For later use, observe that

$$T - q_0 = \frac{(q_0 - 1)(q_1^{(L)} - q_1)}{(q_1 - 1)(q_1^{(L)} - 1)}. \quad (40)$$

Equations (31) and (32) imply $q_0 - 1 \sim n^{-1}$. Lemma 4.3 gives

$$q_1^{(L)} - q_1 = q_1^{(L)}(1 - e^{-\mathcal{D}_n}) \sim \frac{2}{n^2\mathcal{L}_n}.$$

Both $q_1 - 1$ and $q_1^{(L)} - 1$ are asymptotic to n^{-1} . Hence (40) gives

$$T - q_0 \sim \frac{2}{n\mathcal{L}_n},$$

and therefore $A_J \sim n\mathcal{L}_n$.

The second equation in (38), together with $q_0 - 1 \sim n^{-1}$ and $A_J \asymp n \log n$, gives $B_J \sim n$. More precisely, (31) shows

$$q_0 - 1 = \frac{1}{n + \frac{1}{2}} + O\left(\frac{1}{n \log n}\right).$$

It follows that

$$\frac{1}{B_J} = \frac{1}{n + \frac{1}{2}} + O\left(\frac{1}{n \log n}\right),$$

which proves the last estimate in (39). □

Lemma 5.2 (Refined Jacobi scales). *As $n \rightarrow \infty$,*

$$A_J = n\mathcal{L}_n (1 + O(\mathcal{L}_n^{-1})), \quad (41)$$

$$B_J = n + \frac{n}{\mathcal{L}_n} (1 + O(\mathcal{L}_n^{-1})). \quad (42)$$

Consequently, with $b = n + \frac{1}{2}$,

$$B_J - b = \frac{n}{\mathcal{L}_n} (1 + O(\mathcal{L}_n^{-1})).$$

Proof. Keep the notation in the proof of Lemma 5.1. Equations (32), (33), and (30), with one further term absorbed in their relative $O(\mathcal{L}_n^{-1})$ errors, give

$$T - q_0 = \frac{2}{n\mathcal{L}_n} (1 + O(\mathcal{L}_n^{-1})).$$

The first formula in (38) proves (41).

Also,

$$q_0 = \left(1 + \frac{1}{b}\right) e^{-\delta_0},$$

and hence

$$q_0 - 1 = \frac{1}{n} - \frac{2}{n\mathcal{L}_n} + O\left(\frac{1}{n\mathcal{L}_n^2} + \frac{1}{n^2}\right).$$

Using (41) in the second formula of (38) yields

$$\frac{1}{B_J} = \frac{1}{n} - \frac{1}{n\mathcal{L}_n} + O\left(\frac{1}{n\mathcal{L}_n^2} + \frac{1}{n^2}\right).$$

Inversion proves (42). □

Define the Jacobi model ratios

$$R_j^{(J)} := J_j(A_J, B_J, S_J). \quad (43)$$

Lemma 5.3 (Exact matching through R_3). *For $j = 0, 1, 2, 3$,*

$$R_j^{(J)} = R_j.$$

Proof. The choice $S_J = B_J R_1$ matches R_1 . Equations (38) and (36) match q_0 and q_1 . Lemma 2.1 now gives the assertion. □

To determine whether a second positive Jacobi factor can correct R_4 , continue the logarithm of the ratio between the actual coefficients and the first Jacobi model. For z in a fixed neighborhood of $\{0, 1, 2, 3, 4\}$, define

$$\begin{aligned} E_J(z) := & \log \gamma(n+z) - \log \gamma(n) - z \log S_J + z \log A_J \\ & + \log \Gamma(B_J + z) - \log \Gamma(B_J) - \log \Gamma(A_J + z) + \log \Gamma(A_J), \end{aligned} \quad (44)$$

where the branches are real on the positive axis. We write $\psi = \Gamma'/\Gamma$ and $\psi^{(m)}$ for its m th derivative. For integer $j \geq 0$,

$$E_J(j) = \log \frac{R_j}{R_j^{(J)}}.$$

Lemma 5.3 therefore gives

$$E_J(0) = E_J(1) = E_J(2) = E_J(3) = 0. \quad (45)$$

Writing $b = n + \frac{1}{2}$, four differentiations of (44) and (14) give

$$\begin{aligned} E_J^{(4)}(z) = & h^{(4)}(n+z) + \psi^{(3)}(B_J + z) - \psi^{(3)}(b+z) \\ & - \psi^{(3)}(A_J + z). \end{aligned} \quad (46)$$

Lemma 5.4 (Negative Jacobi defect at R_4). *For fixed z in a bounded set,*

$$E_J^{(4)}(z) = -\frac{2}{n^3 \mathcal{L}_n} (1 + O(\mathcal{L}_n^{-1})). \quad (47)$$

In particular,

$$\mathcal{Q}_n := -E_J(4) = \frac{2}{n^3 \mathcal{L}_n} (1 + O(\mathcal{L}_n^{-1})) > 0 \quad (48)$$

for all sufficiently large n .

Proof. Equation (46), together with (23), gives

$$h^{(4)}(n+z) = \frac{4}{n^3 \mathcal{L}_n} (1 + O(\mathcal{L}_n^{-1}))$$

uniformly for bounded z . By Lemma 5.2 and the mean-value formula,

$$\begin{aligned} \psi^{(3)}(B_J + z) - \psi^{(3)}(b + z) &= (B_J - b)\psi^{(4)}(b + z) + O\left((B_J - b)^2 \sup_{u \asymp n} |\psi^{(5)}(u)|\right) \\ &= -\frac{6}{n^3 \mathcal{L}_n} (1 + O(\mathcal{L}_n^{-1})). \end{aligned}$$

Here we used

$$\psi^{(4)}(u) = -\frac{6}{u^4} + O(u^{-5}), \quad \psi^{(5)}(u) = O(u^{-5}).$$

Finally, $\psi^{(3)}(A_J + z) = O(n^{-3} \mathcal{L}_n^{-3})$. This proves (47).

Since E_J vanishes at $0, 1, 2, 3$ by (45), the Hermite–Genocchi formula writes $E_J(4)$ as the average of $E_J^{(4)}$ over the simplex spanned by $0, 1, 2, 3, 4$. The same uniform asymptotic therefore proves (48). $\square$

6 Second correction: the four-coefficient finite-free model

We now use the sign in Lemma 5.4 to match R_4 . For $U, V > 0$, define

$$\mathfrak{q}_k(U, V) := \frac{(U+k)(V+k+1)}{(V+k)(U+k+1)}. \quad (49)$$

This is the k th quotient invariant of the Jacobi family $J_j(U, V, S)$. Since

$$F_j(A, B, C, D, S) = J_j(A, B, S)J_j(C, D, D),$$

the k th quotient invariant of the finite-free family is

$$\mathfrak{q}_k(A, B)\mathfrak{q}_k(C, D).$$

For the first Jacobi model,

$$q_k = \mathfrak{q}_k(A_J, B_J) \quad (k = 0, 1).$$

Moreover, because the Jacobi model agrees with R_j for $0 \leq j \leq 3$,

$$\log \frac{q_2}{\mathfrak{q}_2(A_J, B_J)} = -E_J(4) = \mathcal{Q}_n. \quad (50)$$

Fix

$$D := \frac{1}{2}B_J. \quad (51)$$

This convenient choice places the second Jacobi factor on the same n -scale as the first and leaves three unknown parameters for the three quotient equations. By Lemma 2.1, matching through R_4 now reduces to finding $A, B, C > 0$ satisfying

$$\mathfrak{q}_k(A, B)\mathfrak{q}_k(C, D) = q_k, \quad k = 0, 1, 2. \quad (52)$$

Lemma 6.1 (Four-coefficient parameter matching). *There is an absolute constant $K_r \geq 32$ such that, for all sufficiently large n , system (52) has a real solution with*

$$A = \frac{6}{5}A_J (1 + O(\mathcal{L}_n^{-1})), \quad (53)$$

$$B - B_J = \frac{n}{3\mathcal{L}_n} (1 + O(\mathcal{L}_n^{-1})), \quad (54)$$

$$C - D = \frac{n}{24\mathcal{L}_n} (1 + O(\mathcal{L}_n^{-1})). \quad (55)$$

If

$$n^3 \log^2(n+2) \geq K_4 d^5 \quad (56)$$

with K_4 sufficiently large, then

$$A \geq 8B, \quad B \geq K_r d, \quad D \geq K_r d, \quad 4d \leq C - D \leq \frac{1}{4}D. \quad (57)$$

Proof. Put

$$f_k(U) := \log \frac{U+k}{U+k+1}.$$

After taking logarithms, (52) becomes

$$f_k(A) - f_k(B) + f_k(C) - f_k(D) = \log q_k. \quad (58)$$

Set $L = \mathcal{L}_n$ and introduce scaled variables by

$$A = aA_J, \quad B = B_J + \frac{bn}{L}, \quad C = D + \frac{cn}{L}.$$

The expansion

$$f_k(U) = -\frac{1}{U} + \frac{k + \frac{1}{2}}{U^2} - \frac{k^2 + k + \frac{1}{3}}{U^3} + O(U^{-4}) \quad (59)$$

is uniform, together with its first derivatives in the scaled variables, when $k = 0, 1, 2$ and (a, b, c) ranges over a fixed compact subset of $(0, \infty) \times \mathbb{R}^2$.

To make the implicit-function step explicit, subtract the Jacobi model and set

$$H_{n,k}(a, b, c) := f_k(aA_J) - f_k\left(B_J + \frac{bn}{L}\right) + f_k\left(D + \frac{cn}{L}\right) - f_k(D) - f_k(A_J) + f_k(B_J).$$

Equations (58) and (50) are equivalent to

$$H_{n,0} = H_{n,1} = 0, \quad H_{n,2} = \mathcal{Q}_n.$$

The value and the first two finite differences in k occur at successive powers of n^{-1} . Introduce the correspondingly rescaled map

$$\mathcal{G}_n(a, b, c) := (nLH_{n,0}, n^2L(H_{n,1} - H_{n,0}), n^3L(H_{n,2} - 2H_{n,1} + H_{n,0})).$$

By Lemma 5.4, the required target is

$$(0, 0, n^3L\mathcal{Q}_n) = (0, 0, 2 + O(L^{-1})).$$

The expansion (59) and Lemma 5.2 show, uniformly with first derivatives on a neighborhood of the limiting solution, that

$$\mathcal{G}_n(a, b, c) = (1 - a^{-1} - b + 4c, 2b - 16c, -6b + 96c) + O(L^{-1}).$$

Thus the limiting equations are

$$1 - a^{-1} - b + 4c = 0, \quad (60)$$

$$2b - 16c = 0, \quad (61)$$

$$-6b + 96c = 2. \quad (62)$$

The unique solution is

$$x_* := (a_*, b_*, c_*) = \left(\frac{6}{5}, \frac{1}{3}, \frac{1}{24} \right).$$

We now quantify the perturbation argument. Let Φ_n be $\mathcal{G}_n$ minus its required target and let Φ_∞ be the vector of the three limiting left sides in (60)–(62). Thus

$$\Phi_n(a, b, c) = \mathcal{G}_n(a, b, c) - (0, 0, n^3 L \mathcal{Q}_n),$$

$$\Phi_\infty(a, b, c) = (1 - a^{-1} - b + 4c, 2b - 16c, -6b + 96c - 2).$$

Write $\|\cdot\|_\infty$ for the maximum norm and its induced matrix norm, and set

$$\mathcal{B} := \left\{ x \in \mathbb{R}^3 : \|x - x_*\|_\infty \leq \frac{3}{20} \right\}.$$

Define the actual C^1 error on this cube by

$$\varepsilon_n := \sup_{x \in \mathcal{B}} \max \{ \|\Phi_n(x) - \Phi_\infty(x)\|_\infty, \|D\Phi_n(x) - D\Phi_\infty(x)\|_\infty \}. \quad (63)$$

The uniform expansions above say precisely that $\varepsilon_n = O(L^{-1})$. At x_* the limiting Jacobian and its inverse are

$$J := D\Phi_\infty(x_*) = \begin{pmatrix} 25/36 & -1 & 4 \\ 0 & 2 & -16 \\ 0 & -6 & 96 \end{pmatrix}, \quad J^{-1} = \begin{pmatrix} 36/25 & 27/25 & 3/25 \\ 0 & 1 & 1/6 \\ 0 & 1/16 & 1/48 \end{pmatrix}.$$

In particular, $\|J^{-1}\|_\infty = 66/25$. Moreover, if $x = (a, b, c) \in \mathcal{B}$, then

$$\|I - J^{-1}D\Phi_\infty(x)\|_\infty = \frac{36}{25} \left| a^{-2} - \frac{25}{36} \right| \leq \frac{15}{49}. \quad (64)$$

Take n large enough that $\varepsilon_n \leq 1/30$. The map

$$\mathcal{T}_n(x) := x - J^{-1}\Phi_n(x)$$

satisfies, by (63) and (64),

$$\sup_{x \in \mathcal{B}} \|D\mathcal{T}_n(x)\|_\infty \leq \frac{15}{49} + \frac{66}{25}\varepsilon_n \leq \frac{15}{49} + \frac{11}{125} < \frac{2}{5}.$$

Also

$$\|\mathcal{T}_n(x_*) - x_*\|_\infty \leq \frac{66}{25}\varepsilon_n \leq \frac{11}{125}.$$

It follows that, for every $x \in \mathcal{B}$,

$$\|\mathcal{T}_n(x) - x_*\|_\infty \leq \frac{11}{125} + \frac{2}{5} \frac{3}{20} = \frac{37}{250} < \frac{3}{20}.$$

Thus $\mathcal{T}_n$ maps $\mathcal{B}$ into itself and is a contraction. Its unique fixed point x_n solves the exact system, and the contraction estimate gives

$$\|x_n - x_*\|_\infty \leq \frac{(66/25)\varepsilon_n}{1 - 2/5} = \frac{22}{5}\varepsilon_n = O(L^{-1}). \quad (65)$$

This proves (53)–(55) without a qualitative implicit-function argument.

Finally, (56) gives $d = O(n^{3/5} \log^{2/5} n) = o(n/\log n)$. Since $A/B \sim (6/5)\mathcal{L}_n$, we also have $A \geq 8B$ beyond a fixed threshold. Equations (53)–(55), after increasing K_4 and excluding a fixed finite range of n , imply (57). The finite range is absorbed by one further increase of the eventual absolute constant. $\square$

Put

$$S := BR_1 \quad (66)$$

and define the model ratios

$$R_j^{(\text{F})} := \text{F}_j(A, B, C, D, S). \quad (67)$$

The identity $R_1^{(\text{F})} = S/B = R_1$, together with (52), gives the following.

Lemma 6.2 (Exact matching through R_4). *For $0 \leq j \leq 4$,*

$$R_j^{(\text{F})} = R_j.$$

Proof. The assertion holds at $j = 0, 1$. For the sequence in (67), its three consecutive quotient invariants are precisely

$$\frac{(R_{k+1}^{(\text{F})})^2}{R_k^{(\text{F})} R_{k+2}^{(\text{F})}} = \mathfrak{q}_k(A, B) \mathfrak{q}_k(C, D).$$

Apply (52) successively for $k = 0, 1, 2$. $\square$

Make the change $X = -y/S$ and put

$$P_{\text{F}}(y) := \frac{J^{d,n}(-y/S)}{\gamma(n)}, \quad (68)$$

$$\begin{aligned} p_{\text{F}}(y) &:= \sum_{j=0}^d (-1)^j \binom{d}{j} \frac{(A)_j (C)_j D^j}{A^j C^j (B)_j (D)_j} y^j \\ &= {}_3F_2 \left(\begin{matrix} -d, A, C \\ B, D \end{matrix}; \frac{Dy}{AC} \right). \end{aligned} \quad (69)$$

Define

$$c_j^{(\text{F})} := \frac{R_j}{R_j^{(\text{F})}}. \quad (70)$$

Then P_{F} is obtained from p_{F} by the coefficient multiplier $c_j^{(\text{F})}$, and

$$c_0^{(\text{F})} = \cdots = c_4^{(\text{F})} = 1. \quad (71)$$

7 Real-rootedness and localization of the comparison model

The parameter construction has so far matched coefficients. Its second purpose is algebraic: after the change of variable $X = -y/S$, the finite-free family becomes a coefficientwise convolution of two Jacobi polynomials with positive zeros.

For polynomials written in the normalized form

$$p(y) = \sum_{j=0}^d (-1)^j \binom{d}{j} a_j y^j, \quad q(y) = \sum_{j=0}^d (-1)^j \binom{d}{j} b_j y^j,$$

their multiplicative finite-free convolution is

$$(p \boxtimes_d q)(y) := \sum_{j=0}^d (-1)^j \binom{d}{j} a_j b_j y^j.$$

This operation preserves positive real-rootedness [4, Proposition 2.7]. Its logarithmic-mesh property preserves simplicity when one factor has distinct positive roots [4, Proposition 2.17].

Lemma 7.1 (Multiplicative interval bound). *Suppose the roots of p lie in $[u_-, u_+] \subset (0, \infty)$ and the roots of q lie in $[v_-, v_+] \subset (0, \infty)$. Then every root of $p \boxtimes_d q$ lies in*

$$[u_- v_-, u_+ v_+].$$

Proof. For a polynomial with nonzero constant term, let

$$p^\vee(x) := \frac{x^d p(1/x)}{p(0)}.$$

A coefficient calculation shows that $(p \boxtimes_d q)^\vee = p^\vee \times_d q^\vee$, where $\times_d$ is the monic normalization of the same multiplicative convolution. The largest-root inequality

$$\lambda_{\max}(f \times_d g) \leq \lambda_{\max}(f) \lambda_{\max}(g)$$

is [5, Theorem 1.13]. Applied to $p^\vee, q^\vee$, it gives the lower endpoint $u_- v_-$. Applying it once more after taking reciprocal polynomials gives the upper endpoint $u_+ v_+$. $\square$

Lemma 7.2 (Ordered multiplicative bound). *Let*

$$0 < u_1 \leq \dots \leq u_d, \quad 0 < w_1 \leq \dots \leq w_d$$

be the ordered roots of p and $p \boxtimes_d q$, respectively. If all roots of q belong to $[v_-, v_+] \subset (0, \infty)$, then

$$v_- u_i \leq w_i \leq v_+ u_i \quad (1 \leq i \leq d). \quad (72)$$

Proof. We first recall a consequence of preservation of interlacing. Give positive-rooted monic polynomials the coordinatewise root order. If f precedes g in this order, one can pass from f to g by moving the roots, one at a time, from the largest to the smallest. Each consecutive pair in this finite chain interlaces. Multiplicative finite-free convolution with a positive-rooted polynomial preserves the direction of interlacing [4, Proposition 2.11]; hence it also preserves the coordinatewise root order.

Apply this observation to the reciprocal polynomials used in the proof of Lemma 7.1. The roots of $q^\vee$ lie in $[v_+^{-1}, v_-^{-1}]$, so monotonicity and the identity

$$f \times_d (x - a)^d = a^d f(x/a)$$

give

$$v_+^{-1} \lambda_i(p^\vee) \leq \lambda_i((p \boxtimes_d q)^\vee) \leq v_-^{-1} \lambda_i(p^\vee).$$

Reciprocating and reversing the indices proves (72). $\square$

Lemma 7.3 (Finite-free model roots). *Under (57), the polynomial p_F has d distinct positive roots. Every root and every critical point y satisfies*

$$|y - B| \leq C_{\text{loc}} \sqrt{Bd}, \quad \frac{1}{4}B \leq y \leq 2B \quad (73)$$

with an absolute constant C_{loc} .

Proof. Factor (69) as

$$p_F = p_1 \boxtimes_d p_2, \quad (74)$$

where

$$p_1(y) = {}_2F_1\left(-d, A; B; \frac{y}{A}\right), \quad p_2(y) = {}_2F_1\left(-d, C; D; \frac{Dy}{C}\right).$$

For $U, V > 0$, write

$$q_{U,V}(y) := {}_2F_1\left(-d, U; V; \frac{y}{U}\right) = \frac{d!}{(V)_d} P_d^{(V-1, U-V-d)}\left(1 - \frac{2y}{U}\right).$$

Put $\alpha = V - 1$, $\beta = U - V - d$, and $H = \alpha + \beta = U - d - 1$. After transporting the standard Jacobi matrix [8, Chapter IV] by $y = U(1 - t)/2$, its k th diagonal entry is

$$U \frac{H(V + 2k) + 2k(k + 1)}{(H + 2k)(H + 2k + 2)}. \quad (75)$$

Its off-diagonal entry between rows $k - 1$ and k is

$$\frac{U}{2k + H} \left\{ \frac{k(k + \alpha)(k + \beta)(k + H)}{(2k + H - 1)(2k + H + 1)} \right\}^{1/2}, \quad 1 \leq k < d. \quad (76)$$

We record a ratio-free estimate for this matrix. Suppose that

$$U \geq V + d, \quad V \geq 32d. \quad (77)$$

Then $H \geq V - 1 \geq 31d$. Subtracting V from (75) gives the exact expression

$$\frac{2k(k + H + 1)(U - 2V) + VH(d - 1)}{(H + 2k)(H + 2k + 2)}. \quad (78)$$

Since $|U - 2V| \leq U$, $U = H + d + 1$, and $V \leq H + 1$, the two terms in the absolute value of (78) are bounded respectively by

$$2d \frac{32}{31} \frac{33}{31} \quad \text{and} \quad \frac{32}{31} d.$$

Here we used $k + H + 1 \leq H + d \leq (32/31)H$, $U \leq H + 2d \leq (33/31)H$, and $V \leq H + d \leq (32/31)H$. Their sum is less than $4d$. For an off-diagonal entry put $\beta = U - V - d = H + 1 - V$. Under (77), $0 \leq \beta \leq H$, and

$$\frac{(k + \beta)(k + H)}{(2k + H - 1)(2k + H + 1)} \leq 1.$$

Also $U/(2k + H) \leq 33/31$ and $k + V - 1 \leq V + d \leq (33/32)V$. Hence every off-diagonal entry is at most $2\sqrt{Vd}$, and the two entries adjacent to a row have sum at most $4\sqrt{Vd}$.

Gershgorin's theorem therefore gives

$$\text{roots}(q_{U,V}) \subset [V - C_J \sqrt{Vd}, V + C_J \sqrt{Vd}] \quad (79)$$

with, for example, $C_J = 8$. Indeed, $A \geq 8B$, $B \geq K_r d$, and $K_r \geq 32$ imply (77) for $(U, V) = (A, B)$, so this applies to $p_1 = q_{A,B}$.

For the second factor, use

$$p_2(y) = \frac{d!}{(D)_d} P_d^{(D-1, C-D-d)} \left(1 - \frac{2Dy}{C} \right).$$

Both Jacobi parameters exceed -1 . The inequalities $D \geq K_r d$ and $C - D \geq 4d$ imply (77) for $(U, V) = (C, D)$. We need a sharper estimate than (79) for the semicircle limit. Set $G := C - D$ and $H = C - d - 1 = D + G - d - 1$. Under (57),

$$H \geq D, \quad |C - 2D| \leq D, \quad \frac{C}{H} \leq \frac{17}{16}.$$

The exact difference (78), now with $(U, V) = (C, D)$, is bounded in absolute value by $4d$. Indeed, $k + H + 1 \leq H + d \leq (33/32)H$, so its two numerator terms, after division by the denominator, are at most $(66/32)d$ and d .

For the off-diagonal entries, $\beta = G - d$ and

$$k + D - 1 \leq \frac{33}{32}D, \quad k + \beta \leq G, \quad k + H \leq \frac{33}{32}H.$$

Both denominator factors inside the square root in (76) are at least H . It follows that every off-diagonal entry is at most $2\sqrt{Gd}$. Gershgorin's theorem therefore gives

$$\text{roots}(q_{C,D}) \subset [D - 4d - 4\sqrt{Gd}, D + 4d + 4\sqrt{Gd}].$$

Since $p_2(y) = q_{C,D}(Dy)$,

$$\text{roots}(p_2) \subset [1 - \eta, 1 + \eta], \quad \eta := \frac{4(d + \sqrt{(C - D)d})}{D} < \frac{1}{2}. \quad (80)$$

The last inequality follows from $d/D \leq 1/32$ and $(C - D)/D \leq 1/4$. Both factors have simple positive roots. The cited positivity and logarithmic-mesh theorems give simplicity and positivity for (74). Lemma 7.1, together with $D \asymp B$, gives $|y - B| \leq C_{\text{loc}}\sqrt{Bd}$ for every root. Critical points lie between consecutive roots. Fixing K_r after C_{loc} makes the last two inequalities in (73) follow from $B \geq K_r d$. $\square$

8 The fifth-order residual multiplier

We have constructed a positive-rooted model whose normalized coefficients agree with those of the Jensen polynomial at $j = 0, 1, 2, 3, 4$. We now continue the logarithm of their ratio away from the integers and use those five zeros to control the entire coefficient multiplier.

Lemma 8.1 (Fifth-order residual bound). *There are absolute constants $K_5, C_{\text{def}} > 0$ with the following property. Suppose $d \geq 5$ and*

$$n^3 \log^2(n + 2) \geq K_5 d^5.$$

Put

$$r_F := K_6 \sqrt{Bd}, \quad \Omega_F := \{z \in \mathbb{C} : \text{dist}(z, [0, d]) \leq 2r_F\},$$

where K_6 is a sufficiently large absolute constant. There is a holomorphic function c_F on a neighborhood of Ω_F such that $c_F(j) = c_j^{(F)}$ for $0 \leq j \leq d$ and

$$\sup_{\Omega_F} |c_F - 1| \leq C_{\text{def}} \frac{d^{5/2}}{n^{3/2} \log(n + 2)}. \quad (81)$$

Proof. On the sector in Proposition 4.1, let $g = \log \gamma$ be the branch real on the positive axis. Define

$$\begin{aligned} E_F(z) := & g(n+z) - g(n) - z \log S + z \log A + z \log C - z \log D \\ & + \log \Gamma(B+z) - \log \Gamma(B) - \log \Gamma(A+z) + \log \Gamma(A) \\ & + \log \Gamma(D+z) - \log \Gamma(D) - \log \Gamma(C+z) + \log \Gamma(C). \end{aligned} \quad (82)$$

At every integer $0 \leq j \leq d$, $e^{E_F(j)} = c_j^{(F)}$.

The fifth-wedge hypothesis and Lemma 6.1 imply

$$A \asymp n \log n, \quad B \asymp C \asymp D \asymp n, \quad d + 2r_F = o(n).$$

After increasing K_5 , the set $n + \Omega_F$ lies in the sectorial neighborhood on which g is defined, and every term in (82) is holomorphic on a neighborhood of Ω_F .

Let $b = n + \frac{1}{2}$. Five differentiations, followed by (14), give

$$\begin{aligned} E_F^{(5)}(z) = & h^{(5)}(n+z) + \psi^{(4)}(B+z) - \psi^{(4)}(b+z) - \psi^{(4)}(A+z) \\ & + \psi^{(4)}(D+z) - \psi^{(4)}(C+z). \end{aligned} \quad (83)$$

Equation (25) bounds the first term by $C_*/(n^4 \log n)$. The parameter asymptotics give

$$|B-b| + |C-D| \leq \frac{C_* n}{\log n}.$$

The series for $\psi^{(5)}$ and the mean-value formula therefore give

$$|\psi^{(4)}(B+z) - \psi^{(4)}(b+z)| + |\psi^{(4)}(D+z) - \psi^{(4)}(C+z)| \leq \frac{C_*}{n^4 \log n}.$$

Finally, $|\psi^{(4)}(A+z)| \leq C_*/A^4$. Consequently

$$\sup_{\Omega_F} |E_F^{(5)}| \leq \frac{C_*}{n^4 \log(n+2)}. \quad (84)$$

By (71), the real value of E_F vanishes at $0, 1, 2, 3, 4$. The set Ω_F is convex, so the Hermite–Genocchi formula and (84) give

$$|E_F(z)| \leq \frac{C_*}{5! n^4 \log(n+2)} |z(z-1)(z-2)(z-3)(z-4)|.$$

Since $r_F \geq d$ and $B \asymp n$,

$$\sup_{\Omega_F} |E_F| \leq C_* \frac{r_F^5}{n^4 \log(n+2)} \leq C_* \frac{d^{5/2}}{n^{3/2} \log(n+2)}.$$

For K_5 large the last expression is at most $1/2$. Taking $c_F = e^{E_F}$ and using $|e^w - 1| \leq 2|w|$ for $|w| \leq 1/2$ proves (81). $\square$

9 Critical-point derivative bounds

The multiplier estimate in Lemma 8.1 verifies the analytic hypothesis of Proposition 2.2. It remains to verify the derivative-ratio hypothesis at the critical points of the finite-free model.

Lemma 9.1 (Critical-point derivative ratios). *Let the parameters be the solution supplied by Lemma 6.1, and assume (56) with its constant sufficiently large. Let y be a critical point of p_F and put*

$$T_k := \frac{y^k p_F^{(k)}(y)}{p_F(y)}.$$

If K_6 in the definition of r_F is sufficiently large, then, for all sufficiently large n ,

$$|T_k| \leq r_F^k \quad (0 \leq k \leq d). \quad (85)$$

Proof. The hypergeometric equation is a small perturbation of the Jacobi equation because $(C - D)/C = O(\mathcal{L}_n^{-1})$. We first control the failure of p_F to satisfy the Jacobi equation and then use the differentiated Jacobi recurrence to bound all the ratios T_k .

Write $\mathcal{E} = y \, d/dy$ for the Euler operator and

$$\mathcal{J}p := y \left(1 - \frac{y}{A}\right) p'' + \left\{B - y + \frac{(d-1)y}{A}\right\} p' + dp.$$

The hypergeometric equation for (69) is

$$\left[\mathcal{E}(\mathcal{E} + B - 1)(\mathcal{E} + D - 1) - \frac{Dy}{AC}(\mathcal{E} - d)(\mathcal{E} + A)(\mathcal{E} + C) \right] p_F = 0.$$

Put $\varepsilon_p = (C - D)/C$. Since $D/C = 1 - \varepsilon_p$ and $(\mathcal{E} + D - 1)y = y(\mathcal{E} + D)$, the equation factors exactly as

$$(\mathcal{E} + D)\mathcal{J}p_F = -\frac{\varepsilon_p}{A}\mathcal{E}(\mathcal{E} - d)(\mathcal{E} + A)p_F. \quad (86)$$

Set $W = \mathcal{J}p_F$ and, at the fixed critical point,

$$U_m := \frac{y^m W^{(m)}(y)}{p_F(y)}.$$

Differentiating (86) m times gives

$$U_{m+1} + (D + m)U_m = -\varepsilon_p V_m, \quad (87)$$

where

$$V_m := \frac{y^m}{A p_F(y)} \{ \mathcal{E}(\mathcal{E} - d)(\mathcal{E} + A)p_F \}^{(m)}(y).$$

Since W has degree at most d , $U_{d+1} = 0$.

Write $r = r_F$ and let

$$\mathcal{M} := \max_{0 \leq k \leq d} \frac{|T_k|}{r^k}.$$

Expanding the cubic Euler operator and differentiating monomials gives, with $T_k = 0$ for $k > d$,

$$V_m = \frac{1}{A} \{ T_{m+3} + (A - d + 3 + 3m)T_{m+2} + \beta_m T_{m+1} + \gamma_m T_m \},$$

where the coefficients are explicitly

$$\beta_m = A(2m + 1 - d) - d(2m + 1) + 3m^2 + 3m + 1, \quad \gamma_m = m(m - d)(m + A).$$

For $0 \leq m \leq d$, the bounds in (57) imply $|\beta_m| \leq C_* A d$ and $|\gamma_m| \leq C_* A d^2$. Consequently

$$|V_m| \leq C_* \mathcal{M} r^{m+2} \left(1 + \frac{r}{A} + \frac{d}{A} + \frac{d}{r} + \frac{d^2}{r^2} \right). \quad (88)$$

If $r/D \leq 1/4$, backward substitution in (87), beginning with $U_{d+1} = 0$, gives

$$|U_m| \leq \frac{C_* \varepsilon_p}{D(1-r/D)} \mathcal{M} r^{m+2} \left(1 + \frac{r}{A} + \frac{d}{A} + \frac{d}{r} + \frac{d^2}{r^2} \right). \quad (89)$$

On the other hand, differentiating the Jacobi operator gives the exact identity

$$\begin{aligned} yU_m &= \left(1 - \frac{y}{A} \right) T_{m+2} + \left\{ B + m - y + \frac{(d-1-2m)y}{A} \right\} T_{m+1} \\ &\quad + (d-m) \left(1 + \frac{m}{A} \right) y T_m. \end{aligned} \quad (90)$$

Since $y \leq 2B$ and $A \geq 8B$, we have $1 - y/A \geq 3/4$. For $0 \leq m \leq d-2$, define

$$\begin{aligned} a_m &:= \frac{|B + m - y + (d-1-2m)y/A|}{(1-y/A)r}, \\ b_m &:= \frac{(d-m)(1+m/A)y}{(1-y/A)r^2}. \end{aligned}$$

Lemma 7.3 and $r = K_6 \sqrt{Bd}$ give the explicit bounds

$$a_m \leq \frac{4}{3} \left(\frac{C_{\text{loc}}}{K_6} + \frac{d}{r} + \frac{6dB}{Ar} \right), \quad (91)$$

$$b_m \leq \frac{8}{3K_6^2} \left(1 + \frac{d}{A} \right). \quad (92)$$

The contribution of yU_m in (90), after division by $(1-y/A)r^{m+2}$, is at most $\vartheta_{n,d}\mathcal{M}$, where

$$\vartheta_{n,d} := \frac{4C_* \varepsilon_p}{3} \frac{y/D}{1-r/D} \left(1 + \frac{r}{A} + \frac{d}{A} + \frac{d}{r} + \frac{d^2}{r^2} \right). \quad (93)$$

The wedge hypothesis implies

$$\frac{d}{n} \leq K_4^{-1/5} n^{-2/5} \log^{2/5}(n+2).$$

Together with the parameter asymptotics, this shows uniformly in the wedge that

$$\frac{r}{A}, \frac{d}{A}, \frac{d}{r}, \frac{d^2}{r^2}, \frac{r}{D} \longrightarrow 0, \quad \varepsilon_p = O(\mathcal{L}_n^{-1}), \quad \frac{y}{D} = O(1).$$

Choose K_6 so that the fixed terms in (91)–(92) are small, and then choose n large enough that

$$a_m \leq \frac{1}{4}, \quad b_m \leq \frac{1}{4}, \quad \vartheta_{n,d} \leq \frac{1}{4}$$

for every admissible d and m . Equations (89) and (90) now show

$$\frac{|T_{m+2}|}{r^{m+2}} \leq \frac{1}{4} \frac{|T_{m+1}|}{r^{m+1}} + \frac{1}{4} \frac{|T_m|}{r^m} + \frac{1}{4} \mathcal{M}.$$

Now $T_0 = 1$ and $T_1 = 0$. If $\mathcal{M} > 1$, choose an index at which the maximum is attained. It is at least 2, and the last inequality gives $\mathcal{M} \leq 3\mathcal{M}/4$, a contradiction. Thus $\mathcal{M} \leq 1$, proving (85). $\square$

10 Completion of the proof

Proof of Theorem 1.1. Let n_0 exceed all fixed thresholds in the preceding lemmas. Choose K larger than K_4, K_5 , the constants required in the root and derivative-ratio estimates, and

$$1 + \max_{0 \leq m < n_0} m^3 \log^2(m+2).$$

Increase it once more so that (81) is at most $1/2$ whenever the main hypothesis holds. Then $n^3 \log^2(n+2) \geq Kd^5$ implies $n \geq n_0$.

If $d \leq 4$, then Lemma 6.2 matches every coefficient of the two degree- d polynomials, so $P_F = p_F$. Lemma 7.3 proves the conclusion directly.

Suppose $d \geq 5$. Lemma 7.3 gives d simple positive zeros for p_F . Lemma 8.1 supplies a holomorphic multiplier c_F on Ω_F with

$$\sup_{\Omega_F} |c_F - 1| \leq \frac{1}{2}.$$

At the integers, $c_F(j) = R_j/R_j^{(F)} > 0$, and (71) gives $c_F(0) = \cdots = c_F(4) = 1$. Lemma 9.1 verifies (6) with $r = r_F$. All the hypotheses of Proposition 2.2 are therefore satisfied, so P_F has d simple positive zeros.

Finally,

$$P_F(y) = \frac{J^{d,n}(-y/S)}{\gamma(n)}$$

with $S > 0$. Hence $J^{d,n}$ has d distinct negative real zeros. □

11 The joint semicircle limit

Let μ_{sc} denote the probability measure

$$d\mu_{sc}(x) := \frac{1}{2\pi} \sqrt{4 - x^2} \mathbf{1}_{[-2,2]}(x) dx.$$

Lemma 11.1 (Semicircle law for the comparison model). *Suppose that $n, d \rightarrow \infty$ through pairs satisfying the hypothesis of Theorem 1.1. If $0 < y_1 \leq \cdots \leq y_d$ are the roots of p_F , then*

$$\frac{1}{d} \sum_{i=1}^d \delta_{(y_i - B)/\sqrt{Bd}} \implies \mu_{sc}. \quad (94)$$

Proof. The parameter asymptotics in Lemmas 5.2 and 6.1 give, with $G := C - D$,

$$\frac{B}{n} \longrightarrow 1, \quad \frac{D}{n} \longrightarrow \frac{1}{2}, \quad \frac{B}{A} \longrightarrow 0, \quad \frac{G}{B} \longrightarrow 0. \quad (95)$$

The wedge gives $d/B \rightarrow 0$.

First consider the roots $u_1 \leq \cdots \leq u_d$ of $p_1 = q_{A,B}$. Let $\mathcal{J}_1$ be the Jacobi matrix with entries (75) and (76), for $(U, V) = (A, B)$, and write its diagonal and off-diagonal entries as $D_k^{(1)}$ and $E_k^{(1)}$. The estimate proved before (79) gives

$$\max_{0 \leq k < d} \frac{|D_k^{(1)} - B|}{\sqrt{Bd}} \leq 4\sqrt{\frac{d}{B}} = o(1). \quad (96)$$

If $H = A - d - 1$, the exact off-diagonal formula gives, uniformly for $1 \leq k < d$,

$$\begin{aligned} \frac{(E_k^{(1)})^2}{Bk} &= \frac{A^2}{(H+2k)^2} \frac{B+k-1}{B} \frac{(H+k+1-B)(H+k)}{(H+2k-1)(H+2k+1)} \\ &= 1 + O\left(\frac{d}{B} + \frac{B}{A}\right) = 1 + o(1). \end{aligned} \quad (97)$$

Here $A/(H+2k) = 1 + O(d/A)$, $(B+k-1)/B = 1 + O(d/B)$, and the final quotient on the first line is $1 + O(B/A + d/A)$, all uniformly in k .

Let $\mathcal{H}_d$ be the $d \times d$ tridiagonal matrix with zero diagonal and off-diagonal entries $\sqrt{k}$ between rows $k-1$ and k , $1 \leq k < d$. Equations (96) and (97) imply

$$\left\| \frac{\mathcal{J}_1 - BI}{\sqrt{Bd}} - \frac{\mathcal{H}_d}{\sqrt{d}} \right\|_{\text{op}} = o(1). \quad (98)$$

We used here the elementary bound that the norm of a symmetric tridiagonal matrix is at most the largest absolute diagonal entry plus twice the largest absolute off-diagonal entry. The characteristic polynomial of $\mathcal{H}_d$ is the monic probabilists' Hermite polynomial. Its normalized empirical spectral measure tends to μ_{sc} . For completeness, this follows directly from traces: a nearest-neighbor walk count gives, for every fixed $m \geq 0$,

$$\lim_{d \rightarrow \infty} \frac{1}{d} \text{tr} \left(\frac{\mathcal{H}_d}{\sqrt{d}} \right)^{2m} = \frac{1}{m+1} \binom{2m}{m}, \quad \text{tr}(\mathcal{H}_d^{2m+1}) = 0.$$

Indeed, away from the two boundary rows, each closed walk of length $2m$ has m upward and m downward steps and contributes $(k/d)^m + O_m(d^{-1})$ when it starts in row k ; averaging over k produces $\binom{2m}{m} \int_0^1 x^m dx$. The matrices $\mathcal{H}_d/\sqrt{d}$ have norm at most 2, so the moment limits give weak convergence. Weyl's inequality and (98) now show that

$$\frac{1}{d} \sum_{i=1}^d \delta_{(u_i - B)/\sqrt{Bd}} \implies \mu_{\text{sc}}. \quad (99)$$

It remains to check that the second finite-free factor is negligible on this scale. By (80) and Lemma 7.2,

$$|y_i - u_i| \leq \eta u_i, \quad \eta = \frac{4(d + \sqrt{Gd})}{D}.$$

Equation (79) gives $u_i \leq B + C_J \sqrt{Bd}$. Since B/D stays bounded,

$$\begin{aligned} \max_{1 \leq i \leq d} \frac{|y_i - u_i|}{\sqrt{Bd}} &\ll \frac{d + \sqrt{Gd}}{D} \frac{B + \sqrt{Bd}}{\sqrt{Bd}} \\ &\ll \left(\sqrt{\frac{d}{B}} + \sqrt{\frac{G}{B}} \right) \left(1 + \sqrt{\frac{d}{B}} \right) = o(1) \end{aligned}$$

by (95). Combining this uniform pairing with (99) proves (94). $\square$

Proof of Theorem 1.2. Suppress the index ν . Let $0 < \hat{y}_1 \leq \dots \leq \hat{y}_d$ be the roots of P_F . The proof of Theorem 1.1 applies Proposition 2.2 to p_F and P_F . Consequently, (7) gives

$$\sup_{t > 0} \left| \frac{1}{d} \# \{i : \hat{y}_i \leq t\} - \frac{1}{d} \# \{i : y_i \leq t\} \right| \leq \frac{1}{d}. \quad (100)$$

The same bound holds after any one-to-one affine change of variable.

If ρ_i is the Jensen zero corresponding to $\widehat{y}_i$, then $S = BR_1$ and (68) give

$$\rho_i = -\frac{\widehat{y}_i}{BR_1}, \quad \sqrt{\frac{n}{d}}(1 + R_1\rho_i) = -\sqrt{\frac{n}{B}} \frac{\widehat{y}_i - B}{\sqrt{Bd}}.$$

Lemma 11.1, (100), and $B/n \rightarrow 1$ show that the empirical measure of the expressions on the left converges to the reflection of μ_{sc} . The semicircle law is symmetric, so this reflection is μ_{sc} itself. $\square$

12 Discussion

The published theorem of Griffin, Ono, Rolen, Thorner, Tripp, and Wagner gives hyperbolicity for $n \geq ce^d$; arXiv v3 records the sharper threshold $n \geq ce^{d/2}$ [3, Theorem 1.1]. Their comparison is Hermite. Here the comparison stays in the positive-root class: the positive error at R_3 selects the first Jacobi deformation, and the negative error at R_4 is absorbed by the second finite-free factor.

Theorem 1.2 improves the order of limits in the global derivative-aspect GUE statement of [2]. In their notation $\delta(n) \sim (2n)^{-1/2}$, so the factor $\sqrt{n/d}$ in Theorem 1.2 is the leading form of $1/(\sqrt{2d}\delta(n))$; centering at $-1/R_1$ uses the exact first coefficient ratio. Their theorem first fixes d and lets $n \rightarrow \infty$, obtaining a Hermite polynomial, after which the semicircle law is recovered as $d \rightarrow \infty$. Here the two limits are simultaneous throughout the polynomial wedge. No claim is made about local GUE correlations.

After $R_1, \dots, R_4$ have been matched, the logarithmic multiplier vanishes at five consecutive indices and has fifth derivative $O(1/(n^4 \log n))$. The root localization and perturbed Jacobi equation give the derivative scale $r_F \asymp \sqrt{nd}$. Consequently the remaining error is

$$O\left(\frac{(\sqrt{nd})^5}{n^4 \log n}\right) = O\left(\frac{d^{5/2}}{n^{3/2} \log n}\right),$$

which yields the stated wedge. The constants in the Jacobi localization can be taken explicitly; the proof above uses $C_J = 8$, and the localization itself requires only $A \geq B + d$ rather than $A \geq 64(B + d)$. The later differential recurrence is where the modest condition $A \geq 8B$ is used. No attempt is made to optimize the eventual absolute constant.

This result controls an asymptotic region and does not provide a converse route from partial Jensen hyperbolicity to the Riemann hypothesis; see Farmer [1]. O’Sullivan’s modified Pólya–Jensen criterion gives a complementary use of Hermite combinations [6].

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